# WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY FOR SELF-IMPROVING AGENTS IN ADAPTIVE WORLDS

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## ABSTRACT

Self-improving agents repeatedly replace an incumbent policy with a candidate that a verifier declares better. This loop silently assumes that a gain in the verifier's world remains a gain after deployment changes the world and other agents respond. We define *Improvement Fidelity* for the update transition $\tau = (\pi_t, \pi_{t+1})$, rather than for isolated policy values. The proxy improvement is $\Delta_V = J_V(\pi') - J_V(\pi)$ and the deployment improvement is $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An *improvement reversal* occurs when $\Delta_V > 0$ but $\Delta_* < 0$.

PIVOT (Paired Interventional Validation of Optimization Transitions) is a budgeted validator for this estimand. It pairs incumbent and candidate rollouts under common randomness, models the differential correction $\Delta_* - \Delta_V$, and allocates high-fidelity interventions by expected value of sample information per cost. Six elementary results characterize when global value accuracy is sufficient, why it is not necessary, how update operator shift and environment response can amplify error, and when a selection decision is preserved.

On frozen, preregistered evidence, operator shift increases local error while global policy rank remains competitive; repeated validation changes some closed-loop trajectories; the registered OOD learner comparison is a powered null; and independent adaptive opponents add a negative strategic layer. A public finance audit is explicitly observational and negative for causal claims. We therefore make no universal superiority, simulator-ground-truth, market-causality, or equilibrium claim. The contribution is an estimand and a reproducible decision rule for testing whether learning progress transfers.

## 1 INTRODUCTION

Self-improvement is usually written as a loop: generate a candidate, score it, promote it, and repeat. A verifier can rank policies correctly while ranking improvements incorrectly. The object consumed by that loop is not a policy in isolation but a replacement operation. Deployment can change transitions, rewards, liquidity, or the behavior of other agents, so a verifier can be correct about the world it evaluates and still accept a harmful update. The failure is therefore stronger than a buggy evaluator or a self-authored test: the proxy world is the wrong intervention model.

We study the local transition

$$\tau_t = (\pi_t, \pi_{t+1}) \qquad \text{and not only} \qquad J(\pi_t). \tag{1}$$

Let $V$ be a cheap observer or replay world and let $*$ denote a deployment world that may respond to the deployed policy. Define

$$\Delta_V = J_V(\pi') - J_V(\pi), \qquad \Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi). \tag{2}$$

The right-hand event in fig. 1, $\Delta_V > 0 \wedge \Delta_* < 0$, is the basic failure mode.

Our estimand is operator-relative Improvement Fidelity. An improvement operator $\mathcal{A}$ induces a transition distribution $Q_{\mathcal{A}}$; for a loss $L$,

$$\mathrm{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi,\pi')\sim Q_{\mathcal{A}}} \left[ L(\Delta_V, \Delta_*) \right]. \tag{3}$$

Thus the relevant population is the updates an agent actually proposes, not a uniform population of policies. This distinction motivates three limited contributions:

**Contribution 1.** We formalize Improvement Fidelity and give six propositions covering global sufficiency, a counterexample to the converse, operator shift, response–footprint sensitivity, decision preservation, and finite-sample best-update identification.
**Contribution 2.** We introduce PIVOT-VOI, a paired differential posterior with EVSI-per-cost acquisition and adaptive stopping. PIVOT-H is retained as a transparent heuristic baseline; no universal dominance is claimed.
**Contribution 3.** We release a frozen, hash-indexed evidence contract spanning observer, actor, and strategic worlds, including negative and underpowered outcomes.

## 2 Related Work

**Performative and strategic response.** Performative prediction makes deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2022); recent work studies policy gradients, independent learning, and multi-agent performative dynamics Basu et al. (2025); Sahitaj et al. (2025); Wang et al. (2025); Piliouras & Yu (2022). These works target stable policies, equilibria, or convergence. We instead ask whether a particular replacement remains an improvement after its response is realized.

**World models and simulators.** WorldGym and policy-aware simulator learning evaluate policy values, rankings, or simulator objectives Quevedo et al. (2025); Dann et al. (2026); Hansen et al. (2024). Recent market simulators model endogenous order flow and liquidity with interactive LOB feedback or state–event generation Noble et al. (2026); Zhang et al. (2026); Kawawa-Beaudan et al. (2026). Their primary objects remain policy-level fidelity or trajectory realism. A good global rank can still coexist with a wrong local update sign, which is the gap measured here.

**Self-improvement and verification.** Executable strategy and policy evolution has been studied by EVOQUANT, EvoPolicyGym, Darwin Godel Machine, and Evo-Harness Mao et al. (2026); Wang et al. (2026); Zhang et al. (2025); Wei et al. (2026). Broader agentic software and scientific-design benchmarks include SWE-agent, MLE-bench, OpenHands, and the AI Scientist Yang et al. (2024); Chan et al. (2024); Wang et al. (2024); Lu et al. (2024); AgentBench, WebArena, SWE-bench, tau-bench, and AgentDojo make interactive evaluation executable across task and safety settings Liu et al. (2024); Zhou et al. (2024); Jimenez et al. (2024); Yao et al. (2024); Debenedetti et al. (2024). Self-Authored Verification identifies a verifier–deployment gap when an agent writes its own tests Guo et al. (2026); Progress or Regress? documents reversal in post-training Wu et al. (2024); AI4AI-Bench evaluates recursive algorithm design Chi et al. (2026); and Self-Rewarding Language Models and process verification study feedback signals for iterative improvement Yuan et al. (2024); Lightman et al. (2023). Recent harness optimization and reliability audits further separate capability gain from evaluation stability Kulkarni et al. (2026); Ye et al. (2026). These works motivate executable evaluation but do not define a paired, policy-induced update estimand. Our verifier is fixed and external: it can fail because it evaluates the wrong world, not because the agent edits the test.

**Decision-preserving queries and finance.** Active learning and value-of-information provide the basis for allocating expensive observations Settles (2009); Lindley (1956); paired rollouts additionally cancel common-mode noise. PIVOT changes the target from global prediction loss to update-selection regret. ABIDES-MARL, FinEvo, and FinRL-Meta provide increasingly endogenous financial testbeds Cheridito et al. (2025); Zou et al. (2026); Liu et al. (2022); Byrd et al. (2019); Cont et al. (2010); Nevmyvaka et al. (2006). FinEvo-Bench and recent financial-agent audits measure longitudinal capability, compliance, and execution-interface drift Deng et al. (2026); Li & Zhu (2026). They do not define improvement consistency or a transition-level validator that separates observer, actor, and strategic reversals.

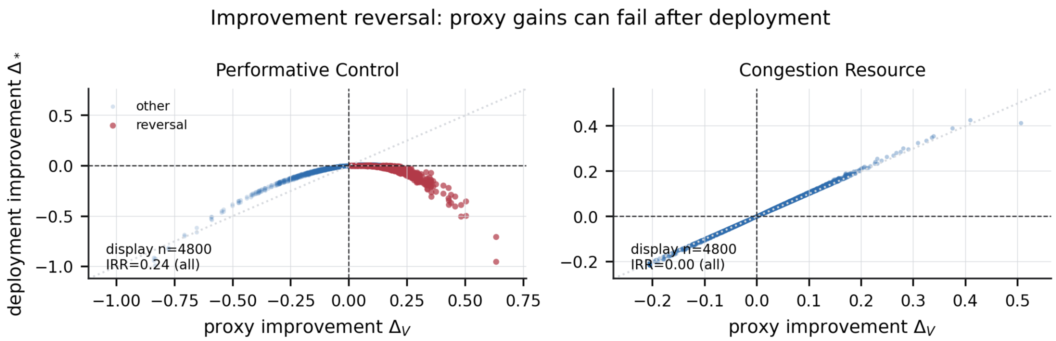


Figure 1: Paired proxy and deployment improvements in the two controlled worlds. The lower-right quadrant is an improvement reversal. Points are a deterministic display subsample; the source bundle retains all transition rows and reports the full-row IRR.

## 3 Problem Formulation

**Policy-induced worlds.** Write deployment response as $\mathcal{M}[\pi]$ and value as

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \tag{4}$$

We use three diagnostic layers:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \tag{5}$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \tag{6}$$

$$\Delta_{\text{strategic}} = J_i(\pi', BR_{-i}(\pi')) - J_i(\pi, BR_{-i}(\pi)). \tag{7}$$

The layers are estimands implemented by separate worlds, not an assertion that a learned or observational proxy is ground truth.

**Metrics and scale.** For transition rows, $\text{IDE} = n^{-1}\sum_k |\Delta_{V,k} - \Delta_{*,k}|$, and ISC is sign agreement after removing ties. The improvement reversal rate is $\text{IRR} = P(\Delta_* < 0 \mid \Delta_V > 0)$; the strategic analogue is $\text{SIRR} = P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$. For a candidate set $C_t$,

$$\text{ISR}_t = \max_{j \in C_t} \Delta_{*,t,j} - \Delta_{*,t,\hat{j}}, \tag{8}$$

and cumulative true improvement is $\text{CTI}_T = \sum_t \Delta_{*,t,\hat{j}}$. We define cumulative improvement-selection regret as $\text{CISR}_T = \sum_t \text{ISR}_t$; the budget experiment has $T = 1$, so its CISR equals ISR for one candidate set. ISR, CTI, and CISR are in native environment reward units, and we never compare their raw magnitudes across environments. When Proxy Only and all-high-fidelity are matched within the same environment, candidate count, and decision horizon, we may report the fraction of excess regret removed,

$$\text{FER} = 1 - \frac{\text{CISR}_m - \text{CISR}_{\text{allHF}}}{\text{CISR}_{\text{proxy}} - \text{CISR}_{\text{allHF}}}; \tag{9}$$

otherwise FER is withheld.

**Value Fidelity versus Improvement Fidelity.** Policy-level value error and update-level error are different estimands. We use "false improvement (improvement reversal)" only for the event $\Delta_V > 0$ and $\Delta_* < 0$.

## 4 Theory

The following results are elementary but delimit what a fidelity claim can mean. Proofs are included to make the assumptions auditable.

**Proposition 1** (Global fidelity is sufficient)**.** *If* $\sup_\pi |J_V(\pi) - J_*(\pi)| \le \varepsilon$, *then* $|\Delta_V - \Delta_*| \le 2\varepsilon$. *The sign is preserved whenever* $|\Delta_*| > 2\varepsilon$.

*Proof.* $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$; the triangle inequality gives the bound and the sign statement. ☐

**Proposition 2** (Global fidelity blindness)**.** *For every $\varepsilon > 0$, there are a finite policy family, a verifier, a policy distribution, and an update operator with policy MAE and rank error below $\varepsilon$, but* $\mathrm{IRR} = 1$ *and* $\mathrm{ISC} = 0$ *on the operator's transition law.*

*Proof.* Take $n$ policies with $J_*(i) = i/(n-1)$ and swap one adjacent pair in $V$. Then $\mathrm{MAE} = 2/[n(n-1)]$ and $1 - \mathrm{Spearman} = 12/[n(n^2-1)]$, both tending to zero. Let the operator sample only the swapped directed transition and propose the lower-valued true policy. Its proxy sign is positive and its deployment sign is negative. ☐

**Proposition 3** (Operator Shift Bound)**.** *Let $P$ be a calibration law, let $Q_{\mathcal{A}} \ll P$ have density ratio $w$, and let $\ell(\tau) = L(\Delta_V, \Delta_*) \geq 0$. Then*

$$\mathrm{IF}(V, \mathcal{A}; L) \leq \sqrt{\mathbb{E}_P[\ell^2]}\sqrt{1 + \chi^2(Q_{\mathcal{A}} \| P)}. \tag{10}$$

*Proof.* Change of measure gives $\mathbb{E}_P[w\ell]$; Cauchy–Schwarz and $\mathbb{E}_P[w^2] = 1 + \chi^2(Q_{\mathcal{A}} \| P)$ complete the proof. ☐

**Proposition 4** (Response–Footprint Bound)**.** *Suppose $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_M d(\pi, \pi')$ and the value functional is $L_J$-Lipschitz in its world argument. Then*

$$|\Delta_{\mathrm{actor}} - \Delta_{\mathrm{direct}}| \leq L_J L_M d(\pi, \pi'). \tag{11}$$

*Proof.* The incumbent term cancels because both comparisons use $\mathcal{M}[\pi]$ as the reference. The candidate term is bounded by $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi])$. ☐

**Proposition 5** (Decision Preservation Under Differential Error)**.** *For candidates $i, j$ with true margin $m = \Delta_{*,i} - \Delta_{*,j} > 0$, if each estimate has absolute error below $m/2$, then the estimated ordering is unchanged.*

*Proof.* Subtract the two estimates. The perturbation of the margin has absolute value below $m$, so the estimated margin remains positive. ☐

**Proposition 6** (Finite-Sample Best-Update Identification)**.** *If paired rollout differences are $\sigma$-sub-Gaussian and the best of $K$ candidates has margin at least $m > 0$, then $n$ paired rollouts per candidate give wrong-selection probability at most $(K-1)\exp[-nm^2/(4\sigma^2)]$. It suffices that $n \geq 4\sigma^2 m^{-2} \log((K-1)/\delta)$ for error at most $\delta$.*

*Proof.* For each competitor, a wrong ordering requires the difference of two sample mean errors to exceed $m$. Apply the sub-Gaussian tail bound and union bound over $K-1$ competitors. ☐

The first two propositions show that global value fidelity is sufficient but not necessary. The shift bound says that local fidelity can deteriorate when the update operator concentrates away from calibration. The response bound motivates measuring an update footprint, while the final two propositions connect differential error to the decision PIVOT must preserve. Strategic responses need not be globally Lipschitz; that is an empirical diagnostic.

# 5 PIVOT

Given an incumbent and $K$ candidates, PIVOT first computes cheap proxy deltas and an update footprint $z_\Delta$. The generic footprint includes policy distance, occupancy shift, action/support shift, and entropy change; the finance footprint includes size, participation, urgency, holding, turnover, liquidity consumption, and spread crossing.

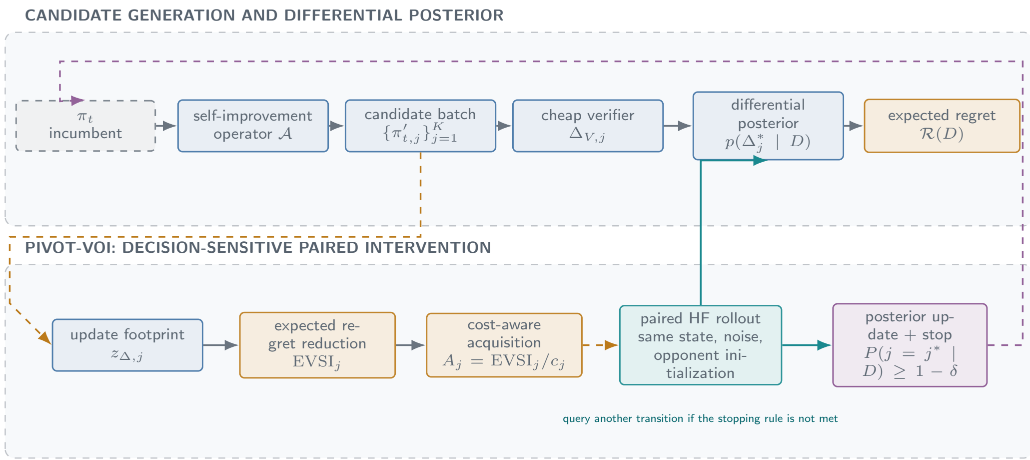


Figure 2: PIVOT's decision-critical path. Candidate generation produces proxy deltas and a differential posterior; footprint-aware EVSI-per-cost selects paired high-fidelity interventions under common state, noise, and opponent initialization. Posterior regret and the stopping probability determine whether another transition is queried or an update is promoted.

**Paired differential correction.** For a queried transition, incumbent and candidate share initial state, exogenous path, random seed, and opponent initialization. PIVOT fits

$$g_\theta(z_\Delta) \approx \Delta_* - \Delta_V, \qquad \widehat{\Delta}_* = \Delta_V + g_\theta(z_\Delta). \tag{12}$$

Pairing removes common-mode noise before aggregation; the unpaired ablation uses disjoint context seeds.

**VOI and stopping.** Let $D$ be current paired evidence and $a(D) = \arg\max_j \mathbb{E}[\Delta_j^* \mid D]$. Define posterior expected simple regret $\mathcal{R}(D) = \mathbb{E}[\max_j \Delta_j^* - \Delta_{a(D)}^* \mid D]$. A hypothetical high-fidelity observation $Y_j$ has

$$\mathrm{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j|D}\mathcal{R}(D \cup \{Y_j\}), \qquad A_j = \mathrm{EVSI}_j/c_j. \tag{13}$$

PIVOT-VOI queries the largest $A_j$ and stops when posterior selection probability exceeds $1 - \delta$ or all acquisition values fall below $\eta$. PIVOT-H, the earlier transparent footprint score, is a baseline. Queried values replace predictions; unqueried candidates retain their proxy/model estimates. The target is update-selection regret, not global world-model error.

**Why Transition Validation Differs from Active Learning.** Active learning usually chooses a label that reduces a global prediction loss. PIVOT chooses a paired intervention that can change the sign or ordering of a replacement operation; its loss is update-selection regret.

## 6 Experiments

**Design and reporting contract.** We use two controlled response worlds, three update operators, ten shift levels, paired common-random-number rollouts, and 30 independent seeds. The unit of inference is a seed, trajectory, or opponent-seed cluster as stated in each figure; transition rows are retained for audit and are not treated as independent replicates. The external adaptive reference and the public finance audit are pinned boundary evidence, not relabeled as controlled ground truth.

**Operator shift.** Across 30 seeds and 36000 paired rows, the high-minus-low shift contrast in operator-relative IDE is 0.2627 with bootstrap interval $[0.1684, 0.3763]$. This supports the local estimand: an operator can move into regions where update error changes even while policy-level rank statistics remain useful. It does not imply that every operator fails.

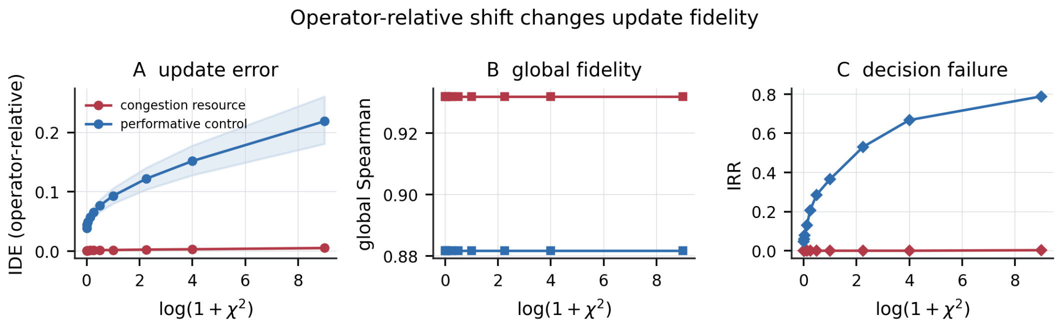


Figure 3: Operator-relative shift diagnostic. Local improvement error rises with the registered shift while global policy rank remains comparatively stable. The three panels show the error, global rank, and reversal rate using the same environment-by-shift cells.

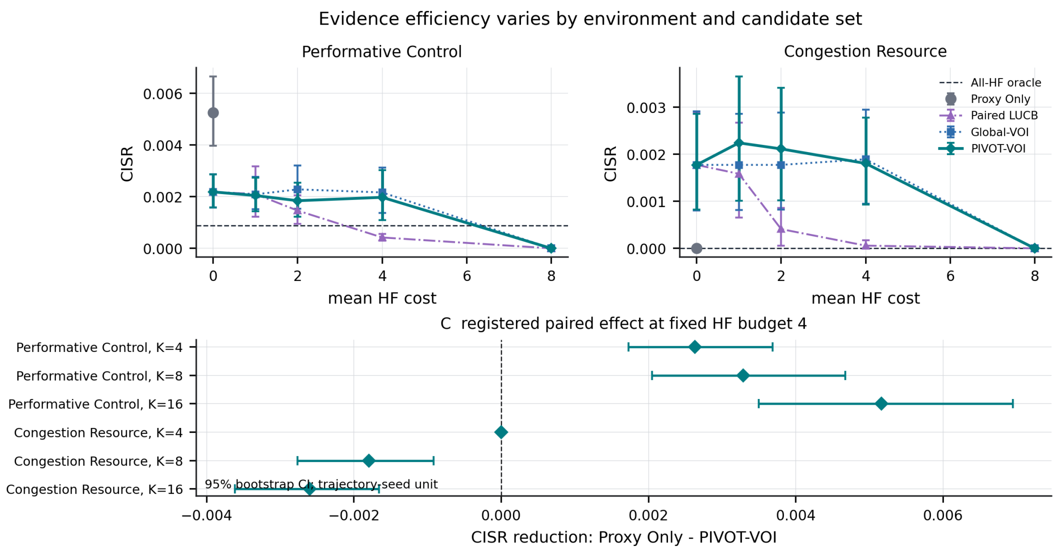


Figure 4: Evidence-efficiency frontier. The top panels fix $K = 8$ and show Proxy Only, Paired LUCB, Global-VOI, and PIVOT-VOI at matched query costs; the all-high-fidelity line is an oracle reference and the proxy point has zero query cost. The lower forest reports the paired CISR reduction of PIVOT-VOI relative to Proxy Only (Proxy Only minus PIVOT-VOI) at budget four over $K \in \{4, 8, 16\}$ and the two worlds. Intervals are seed bootstraps; no uniform method ordering is asserted.

**Closed-loop outcomes.** The repeated loop runs 40 rounds with eight candidates. At the trajectory level, PIVOT-VOI versus Proxy Only has paired CISR reduction 1.4883 with interval [1.0618, 1.9105] (positive favors PIVOT-VOI). The comparison is scoped: the contrast against Global-VOI is negative in this metric and the comparison against Paired LUCB is not uniformly favorable. Figure 5 displays cumulative CTI and CISR; its environment panels use their native reward scales.

**OOD evaluator contrast.** The registered learned transition evaluator is a powered null rather than a win claim. Across 30 seeds and 36000 source rows, transition minus global ISC is -0.2590 with interval [-0.3588, -0.1707]. Appendix Figure 7 shows the split-level forest and paired raw reports. This result separates the need for an update-level estimand from the stronger, unsupported claim that one differential model must dominate every global model.

**Strategic response.** Only a focal policy self-improves; competitors are independent adaptive mechanisms. Across 30 matched opponent-seed clusters, we evaluate the 3 registered adaptive families (best-response, gradient-adaptive, and RL/evolutionary), giving 90 family-by-seed traces before the family-balanced aggregation. The resulting strategic effect is -0.0240 with interval [-0.0249, -0.0231]. The reported SIRR 0.9495 is the unweighted mean of the three family-level SIRRs,

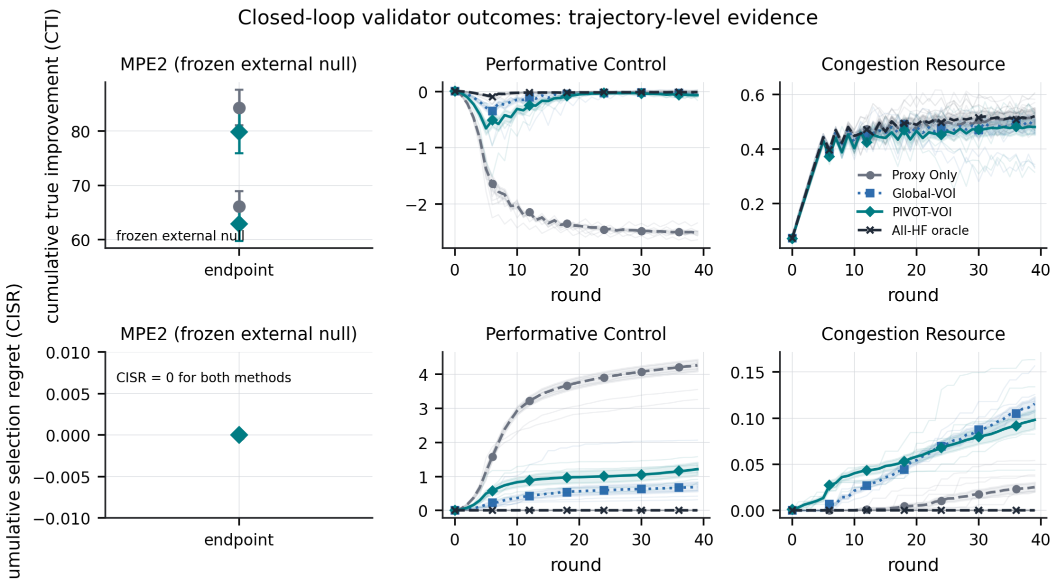


Figure 5: Closed-loop trajectory evidence. Faint traces are independent seeds and ribbons are bootstrap intervals over seeds. The two controlled worlds are reconstructed round by round from paired transition rows; the frozen external reference is endpoint-only and is not connected as a trajectory.

rather than a pooled transition-row rate. In the strongest cases, $\Delta_{\text{actor}} > 0$ while $\Delta_{\text{strategic}} < 0$: a candidate survives its own mechanical response but fails after competitors adapt. Figures 6 and 9 show the response decomposition and opponent-family generalization. These are scoped to the held-out mechanisms, not an equilibrium or market-ecology result.

**Finance audit and boundary.** The public-data stress test uses historical paths, virtual fills, and a participation/depth proxy. Figure 10 visualizes the initial seven-session single-asset participation sweep. A separately frozen three-asset, four-date expansion contains 12 primary sessions: seven are F1-positive, with 0/7 depth-proxy reversals. Its six-session holdout contains five F1-positive sessions and 0/5 reversals. The expanded pooled depth-minus-replay effect is approximately $-4.16 \times 10^{-7}$ in its native session scale. This is a boundary and plausibility diagnostic; it does not identify causal market impact or treat observed paths as an endogenous simulator.

## 7 Stress Tests Beyond Controlled Environments

The evidence supports a narrow implication: a self-improvement evaluator should be audited on the update transitions it induces, and fidelity should be increased where footprint and response sensitivity can change a decision. It does not support the converse implication that every transition learner is better, nor that PIVOT-VOI dominates all active-query rules. The controlled worlds are deliberately transparent; strategic best responses are finite mechanisms rather than a general equilibrium; and the finance extension is observational. LLM/EvoQuant and generative market-model adapters are useful future operators/world proxies, but they are not required to establish the estimand and are not used as hidden evidence here.

## 8 Conclusion

An agent can be correct about the wrong world. The relevant statistical object for self-improvement is therefore the transition $\pi \to \pi'$, and the core question is whether its sign and ordering survive deployment-induced response. Improvement Fidelity makes that question explicit; PIVOT provides a paired, budgeted way to test it. The resulting evidence ladder is

$$\text{observer} \longrightarrow \text{actor} \longrightarrow \text{strategic}, \tag{14}$$

with each arrow an additional intervention, not an assumed truth claim. Valid progress in adaptive worlds means validating consequences, not merely certifying actions.

**AI Use Statement.** Generative AI tools were used for language editing, code and figure-formatting assistance, limited assistance in developing experimental infrastructure, and literature discovery. All experimental designs, mathematical claims, generated code, numerical results, cited sources, and manuscript claims were reviewed and verified by the authors, who take full responsibility for the work.

**Reproducibility Statement.** The anonymous supplement contains deterministic configurations, hash-indexed source evidence, figure tables, and rebuild commands. The public finance audit uses virtual fills and contains no credentials, private data, raw vendor archives, or live orders.

**Ethics Statement.** The study uses controlled virtual environments and public market data only; it involves no human subjects, personal data, live trading, or external authorization.

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# A Extended Figures and Artifact Notes

The release includes the frozen source snapshot, confirmatory result manifests, and generated figure bundles. Every figure bundle contains PDF, SVG, PNG, CSV, Parquet, metadata, source hashes, and the analysis script. The metadata records the unit of inference and distinguishes descriptive spans from bootstrap intervals.

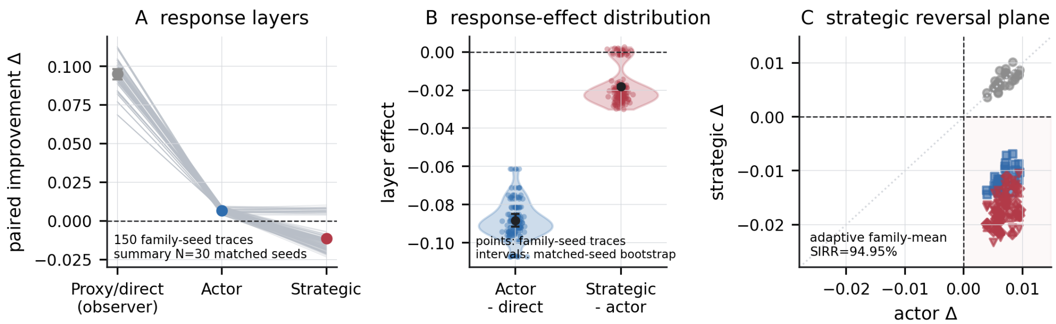


Figure 6: World-response decomposition. Panel A traces 150 family-by-seed observations (five opponent families $\times$ 30 matched seeds) through proxy/direct observer, actor, and strategic worlds; its summary intervals use the 30 matched seeds. Panel B separates mechanical and strategic effects; Panel C marks the strategic reversal quadrant.

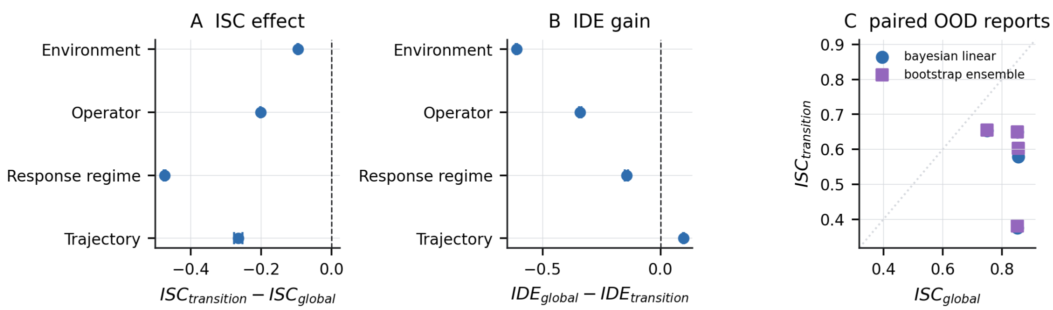


Figure 7: Powered OOD null. Forest horizontal ranges are descriptive spans across the two fitted model families in each split, not confidence intervals; the scatter retains the paired reports and the diagonal indicates equal ISC.

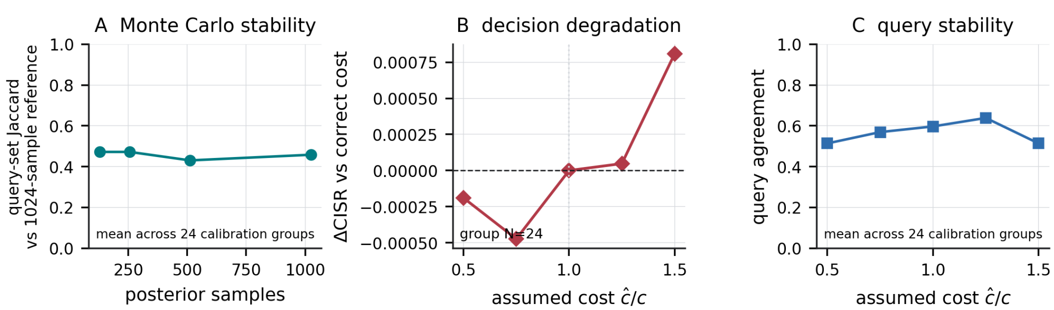


Figure 8: PIVOT-VOI robustness to posterior sample count and cost misspecification. The curves report means over the registered calibration groups; they are numerical-stability diagnostics, not additional superiority claims.

**Metric and claim boundary.** The canonical metric implementation is the frozen statistics module. The closed-loop CTI/CISR values sum over rounds; the budget frontier is a one-set ($T = 1$) selection problem. Their scales are therefore reported separately. The external adaptive and finance artifacts remain bounded evidence; no claim is promoted merely because a plot is visually strong. Internal experiment identifiers, configuration digests, and exact source hashes are retained in the supplement

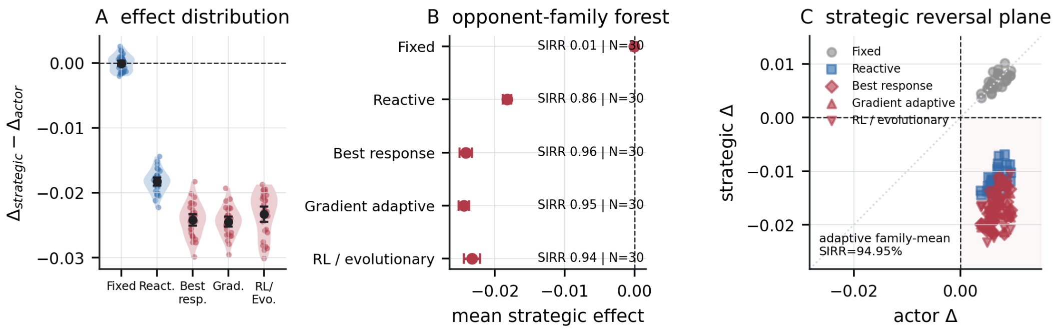


Figure 9: Strategic generalization across opponent mechanisms. Raincloud and forest views show family-specific effects and SIRR, each over 30 matched opponent-seed clusters. The reversal plane labels the unweighted mean over the three adaptive family-level SIRRs; it uses both color and marker shape so the failure is legible in grayscale.

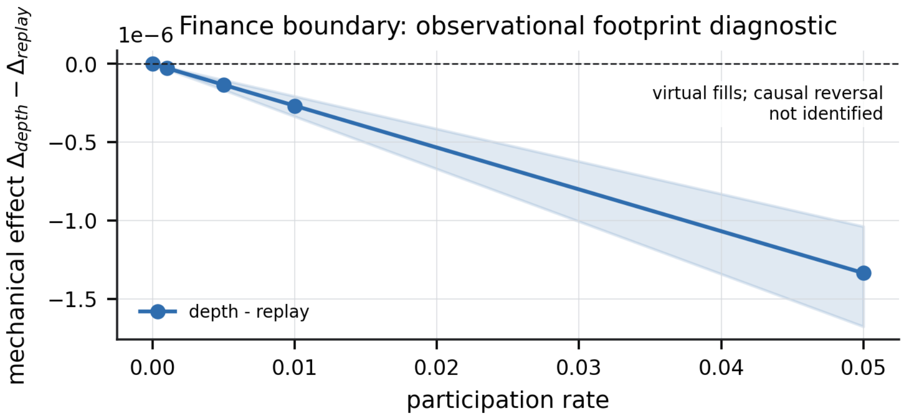


Figure 10: Initial single-asset public finance boundary diagnostic over participation. Shading is a seven-session descriptive interval; fills are virtual and the figure does not identify causal impact. The separate 12-session, three-asset expansion is summarized in the public-data table.

and figure metadata for audit. The raw sampled reversal-rate cells used in the operator-shift display remain available alongside the complete compressed transition stream.

Table 1: Controlled and public evidence summary.

| Evidence layer | Estimate | 95% interval | Unit / interpretation |
|---|---|---|---|
| Operator shift | 0.2627 | [0.1684, 0.3763] | IDE contrast; paired transition |
| Closed-loop validation | 1.4883 | [1.0618, 1.9105] | CISR reduction; trajectory seed |
| OOD evaluator contrast | -0.2590 | [-0.3588, -0.1707] | transition ISC minus global ISC; null |
| Strategic adaptation | -0.0240 | [-0.0249, -0.0231] | strategic minus actor; opponent cluster |

Table 2: Public-data boundary summary.

| Public-data boundary | Value |
|---|---|
| Initial single-asset sessions | 7 |
| Expanded primary sessions | 12 |
| Expanded primary reversals / eligible | 0/7 |
| Expanded holdout reversals / eligible | 0/5 |
| Expanded pooled depth effect | -4.16e-07 |
| Causal impact identified | No |